

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Fundamentals of Data Science

COURSE CODE: DSA04 02

COURSE OUTCOME COVERED

CO1: Analyze the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL2)

QUESTIONS: 20

Total Marks:100

Annexure A

SHORT ANSWER TEST

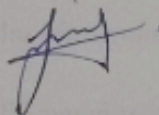
Code of Conduct:

Jannathul Firdaus . N (192573025)

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Short Answer Test

1. A college wants to identify students who may fail in the final exam. How can Data Science help?
2. A hospital has patient records for many years. How can Data Science be applied?
3. An online shopping website wants to recommend products to customers. Which use of Data Science is applied here?
4. A bank wants to detect suspicious transactions. How can Data Science be useful?
5. A company wants to reduce customer loss. How can Data Science support this?
6. A school wants to improve student performance. Mention one benefit of Data Science in this case.
7. A supermarket wants to know which products sell more during festivals. How can Data Science help?
8. A transport department wants to reduce traffic congestion. How can Data Science be applied?
9. A farmer wants to improve crop yield. Give one use of Data Science.
10. A company has data in Excel tables, customer reviews, images, and videos. Which facet of data is shown here?
11. A weather app updates temperature every minute. Which facet of data is involved?
12. A university stores student details in rows and columns. What type of data structure is this?
13. Customer comments on social media are collected for analysis. What type of data is this?
14. A company collects a very large amount of daily user activity data from its mobile app. Which Big Data characteristic is applied?
15. A hospital dataset contains missing age values and incorrect phone numbers. Which data quality issue is present?
16. A business wants to convert customer data into useful sales decisions. Which Big Data characteristic is important?
17. A company stores large files in Hadoop Distributed File System. Which part of the Big Data ecosystem is used?
18. A company wants to process large datasets using parallel computation. Which Big Data tool can be used?
19. An organization wants to collect live sensor data from IoT devices. Which Big Data ecosystem component is suitable?
20. A company wants to visualize Big Data results for decision-makers. Which tool or method can be applied?

CRITERIA FOR CALCULATING

Criteria	Weightage
Conceptual Understanding	
Accuracy	

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

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2.

A college wants to identify students who may fail in the final exam. How can Data Science help?

Data Science can analyze students past performance attendance and exam results to predict which students may be at risk of failing.

2. An hospital has patients online shopping website wants to recommend products to customers. which use of data science is applied here?

This is recommendation systems. Data Science analyzes customer preferences and past purchases to recommend suitable pattern.

3. A company wants to reduce customer loss how can Data Science support this?

Data Science can analyze customer behaviour and identify customers who are likely to leave helping the company takes actions to reduce customer churn.

4. A Supermarket wants to know which products sell more during festivals. How can data science help?

Data Science can analyze historical sales data to identify products that have higher demand during festivals.

5. A farmer wants to improve crop yield. Give one use of data science.

Data Science can analyze weather, soil, rainfall and crop data to predict and improve crop yield.

6. A weather app updates temperature every minutes. which facets of data is involved

"velocity" is involved, because the temperature data is being generated and updated continuously at a high frequency.

7. Customer comments on social media are collected for analysis. What type of data is this?

customer comments are unstructured data mainly in the form of text.

8. A hospital dataset contains missing age values and incorrect phone numbers. which data quality issue is present?

The data has data quality issue such as missing values and inaccurate/incorrect data.

9. A company stores large files in Hadoop distributed file system. which part of the Big data ecosystem is used?

HDFS is the storage component of the Hadoop / Big data ecosystem.

10. An organization wants to collect live sensor data from IoT devices. which Big data ecosystem component is suitable?

Apache Kafka is suitable for collecting and streaming live sensor data from IoT devices.